Euclidean Geometry In Mathematical Olympiads

Lemma 1.48 (Simson Line)

fgm

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illogical_21
1034 posts

Mar 17, 2019, 7:04 PM • 2 #1

Let ABC be a triangle and P be any point on (ABC). Let X, Y, Z be the feet of the perpendiculars from P onto lines $BC, CA,$ and AB. Prove that points X, Y, Z are collinear.

illogical_21
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Mar 17, 2019, 7:05 PM • 2 #2

Solution

Note that $PABC, PAYZ, PBZX, PCXY$ are cyclic, then $\measuredangle PXY = \measuredangle PCY = \measuredangle PBZ = \measuredangle PXZ \implies X, Y, Z$ collinear, so we are done.

kootrapali
4527 posts

Mar 1, 2020, 8:42 PM #3

Solution

Let X and Y be the feet of the perpendiculars from P to BC and AC, respectively, and let $Z' = XY \cap AB$. Since $ABCP$ is cyclic, we have $\measuredangle ABC = \measuredangle APC$.
From $\triangle ABC$ we have
$\measuredangle ABC + \measuredangle BCA + \measuredangle CAB = 0^\circ \rightarrow \measuredangle ABC - \measuredangle ACB - \measuredangle BAC = 0^\circ$.
Plugging this into the first equation yields $(1) \measuredangle APC = \measuredangle ACB + \measuredangle BAC$.
Since $\measuredangle PYC = \measuredangle PXC = 90^\circ$, quadrilateral $PYXC$ is cyclic, so $\measuredangle YPX = \measuredangle YCX = \measuredangle ACB$. Combining this with (1) results in $(2) \measuredangle APC = \measuredangle YPX + \measuredangle BAC$.
Note that
$\measuredangle APY + \measuredangle YPX + \measuredangle XPC = \measuredangle APC \rightarrow \measuredangle APY + \measuredangle YPX + \measuredangle XPC = \measuredangle YPX + \measuredangle BAC$ from (2). From this we get $\measuredangle XPC = \measuredangle BAC - \measuredangle APY$, and from $PYXC$ we get $(3) \measuredangle XYC = \measuredangle BAC - \measuredangle APY$.

We also have
$\measuredangle XYC + \measuredangle CYP + \measuredangle PYZ' = \measuredangle XYZ' = 0^\circ \rightarrow \measuredangle BAC - \measuredangle APY + 90^\circ + \measuredangle PYZ' = 0^\circ$, and this becomes $(4) \measuredangle PYZ' = 90^\circ + \measuredangle APY - \measuredangle BAC$.
From $\triangle APY$, we have
$\measuredangle YAP + \measuredangle APY + \measuredangle PYA = 0^\circ \rightarrow \measuredangle CAP + \measuredangle APY = 90^\circ \rightarrow (5) \measuredangle CAP = 90^\circ - \measuredangle APY$.
Also,
$\measuredangle BAC + \measuredangle CAP + \measuredangle PAZ' = \measuredangle BAZ' = 0^\circ \rightarrow \measuredangle BAC + 90^\circ - \measuredangle APY + \measuredangle PAZ' = 0^\circ \rightarrow (6) \measuredangle PAZ' = 90^\circ + \measuredangle APY - \measuredangle BAC$.
From (4) and (6), we have $\measuredangle PYZ' = \measuredangle PAZ'$, so quadrilateral $PYAZ'$ is cyclic. Thus, $\measuredangle PZ'A = \measuredangle PYA = 90^\circ$, so $PZ' \perp AB$ and $Z = Z'$. Thus, X, Y, Z are collinear.

zuss77
520 posts

Aug 8, 2020, 5:35 AM #5

Solution

$P \in (CXY), P \in (AYZ)$.
$\measuredangle PYZ = \measuredangle PAZ = \measuredangle PAB = \measuredangle PCB = \measuredangle PCX = \measuredangle PYX$
$\implies X, Y, Z$ - collinear.

Aug 14, 2020, 12:33 PM • 3 #6

Jerry122805
1074 posts

Solution (no phantom no directed)

This is a simple angle chase problem. Let $\angle PYZ = x.$ We see $\angle PAZ = x$ by cyclic $PAZY$ and consequentially we know $\angle PAB = 180 - x.$ Then by cyclic $ABCP$ we know $\angle PCB = x$ and $\angle PYX = 180 - x.$ Therefore $\angle PYZ + \angle PYX = 180$ and we are done.

Solution no phantom but directed

We know $\measuredangle PYZ = \measuredangle PAZ = \measuredangle PAB = \measuredangle PCB = \measuredangle PYX$ so ZYX are collinear.

phantom points no directed